

A Complete Möbius and Morse–Bott Atlas for a Symmetric Matrix Riccati Flow

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Abstract

For every finite dimension we solve $\dot{X} = I - X^2$ on real symmetric matrices and classify its maximal classical chart, every signed-time pole, all forward limits, and the exact convergence remainder.

1 Exact flow and maximal chart

Let $X_0 \in \text{Sym}(n, \mathbb{R})$ and consider

$$\dot{X} = I - X^2, \quad X(0) = X_0. \quad (1)$$

Put $C = \cosh t$, $S = \sinh t$.

Theorem 1 (Möbius, pole, and limit atlas). *On the connected component through zero of $\{t : \det(CI + SX_0) \neq 0\}$, the unique classical solution is*

$$F_t(X_0) = (CX_0 + SI)(CI + SX_0)^{-1}. \quad (2)$$

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of X_0 , the only finite chart singularities are

$$t_j = \text{arctanh}(-1/\lambda_j), \quad |\lambda_j| > 1, \quad (3)$$

with their spectral multiplicities. Thus the flow is forward global iff $\lambda_{\min}(X_0) \geq -1$. In that case

$$F_t(X_0) \longrightarrow I - 2P_{\ker(X_0 + I)}. \quad (4)$$

For each $\lambda > -1$ the exact scalar remainder is

$$f_t(\lambda) - 1 = \frac{2(\lambda - 1)e^{-2t}}{(1 + \lambda) + (1 - \lambda)e^{-2t}}. \quad (5)$$

Moreover $F_t \circ F_s = F_{t+s}$ whenever both expressions remain in one classical chart.

Proof. Solve the symmetric block-linear system

$$\dot{U} = V, \quad \dot{V} = U, \quad U(0) = I, \quad V(0) = X_0.$$

It gives $U = CI + SX_0$ and $V = SI + CX_0$. While U is invertible, $X = VU^{-1}$ obeys (1), proving (2); local uniqueness then gives maximality in that chart. The spectral theorem reduces F_t to $f_t(\lambda) = (C\lambda + S)/(C + S\lambda)$. Its denominator vanishes exactly as in (3). For $t > 0$ this happens precisely when $\lambda < -1$. The values -1 and 1 are fixed, every $\lambda > -1$ tends to 1 , and elementary exponential algebra yields (5). Multiplication of the two scalar Möbius matrices gives the addition law, hence the semigroup identity by functional calculus. $\square$

Source lineage

The citation supplies matrix-Riccati and symmetric-flow context, not a literature-priority claim for this packaging.

References

- [1] U. Helmke, “Isospectral flows on symmetric matrices and the Riccati equation,” *Systems & Control Letters* 16 (1991), 159–165. DOI: [10.1016/0167-6911\(91\)90044-F](https://doi.org/10.1016/0167-6911(91)90044-F).